Deepak Mallampati

AI ENGINEER — RAG & DOCUMENT AI

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US-based · F-1 OPT · open to Dallas relocation

PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years building production **retrieval-augmented generation (RAG)** and LLM systems over document-heavy, regulated corpora in banking and enterprise. Delivered **92% retrieval precision** across millions of regulatory documents and **40% hallucination reduction** using embeddings, chunking, vector search, and cross-encoder re-ranking on **Azure OpenAI** and OpenAI APIs. Strong in LLM orchestration (LangChain/LangGraph), evaluation (RAGAS, BERTScore), and compliant cloud deployment — directly transferable to document-AI for commercial real estate.

CORE COMPETENCIES

RAG Architecture • Embeddings & Vector Search • Document Retrieval • Azure OpenAI & OpenAI APIs • LLM Orchestration (LangChain/LangGraph) • Document Chunking & Re-ranking • LLM Evaluation (RAGAS, BERTScore) • Cloud AI Deployment (Azure)

Generative AI & RAG: RAG Architecture | Embeddings, Chunking & Document Retrieval | Cross-Encoder Re-ranking | Semantic & Hybrid Search | Semantic Caching | Prompt Engineering | Multi-Agent Systems

LLMs & Orchestration: Azure OpenAI, GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | Hugging Face Transformers | Fine-tuning (LoRA, QLoRA, PEFT)

Vector Search: FAISS, Pinecone, Weaviate | Document Chunking Strategies

MLOps & Cloud: Azure (OpenAI, AI Services, Content Safety, DevOps), AWS (Bedrock, SageMaker, Lambda), GCP Vertex AI | Docker, Kubernetes | CI/CD | MLflow, Weights & Biases | Model Monitoring

ML & NLP: PyTorch, TensorFlow | NER, Text Classification, Summarization | RAGAS Evaluation | BERTScore | Drift Detection

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | PostgreSQL, MongoDB, Redis

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present*

AI Engineer — USA

- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking and tuned chunking strategies to lift answer relevance and reduce hallucination on domain-specific queries by **40%**.
- Architected production-grade **agentic LLM systems** on a conductor–subagent orchestration framework, decomposing complex objectives into context-scoped subagents — reducing token consumption by **42%** while sustaining task accuracy.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.
- Designed **privacy-preserving data workflows** for sensitive datasets (K-anonymity, L-diversity, differential privacy) enabling compliant analytics with zero PII exposure.

Vanguard *Jan 2024 - Jan 2025*

AI Engineer Intern — USA

- Built **12 APIs and microservices** powering new AI capabilities over Vanguard's internal platform, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini** via Azure OpenAI, improving task success rates by **27%** on internal evaluation datasets and informing model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%**.
- Developed and enhanced AI agents and backend services across **25+ AI agents**, partnering with product, platform, and data engineering to support **100,000+ requests daily** across 20+ internal teams.

Kent State University *Oct 2023 - Jan 2024*

ML & Gen AI Research Assistant — USA

- Built a privacy-preserving ML pipeline on clinical data (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation; authored an IEEE-format conference paper documenting the methodology.

Emerson *Jun 2022 - Apr 2023*

ML Trainee — India

- Developed supervised regression and classification models for equipment-failure prediction, anomaly detection, and capacity forecasting on operational data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific financial datasets, achieving **89% F1** on custom NER tasks.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

A framework that analyzes sensitive records without exposing identities, combining k-anonymity, l-diversity, and differential privacy — quantifying the exact privacy/accuracy trade-off for document-heavy regulated data.

Python · Differential Privacy · scikit-learn

career-ops

An autonomous, AI-driven pipeline that scans listings, scores fit with RAG-style retrieval, and generates tailored documents overnight without supervision.

Python · LLM Orchestration · RAG

MangaLens Chrome Web Store

A shipped Chrome extension delivering real-time AI translation across 29+ sites, turning a multi-step manual chore into a single inline action.

TypeScript · LLM APIs

EDUCATION

M.S. in Computer Science — *Kent State University, OH* *2025 · GPA 3.70/4.0*